



C++ I.

LAB 03.

Chapter 3.

Create a lab report for the following programming challenge.

17. Math Tutor

Write a program that can be used as a math tutor for a young student. The program should display two random numbers to be added, such as

$$\begin{array}{r} 247 \\ +129 \\ \hline \end{array}$$

The program should then pause while the student works on the problem. When the student is ready to check the answer, he or she can press a key and the program will display the correct solution:

$$\begin{array}{r} 247 \\ +129 \\ \hline 376 \end{array}$$

Lab 3: Math Tutor Program - Instructions for Submission

Objective:

Create a C++ program named **tutor.cpp** that acts as a math tutor for young students by generating two random numbers (each less than or equal to 1000) and displaying them for addition. The program should pause while the student solves the problem and display the correct answer when the student is ready.

What You Need to Submit:

You must submit **two files**:

1. **Lab report (PDF):** Named `lab03.pdf`
 2. **C++ source code file:** Named `tutor.cpp`
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Lab Report Requirements (lab03.pdf):

Your lab report must include the following:

- **Proof that your program works correctly:**
Take clear screenshots or photos showing the input numbers displayed and the output answer after pressing a key to reveal it.
- **Run your program at least 5 times:**
Each run should show different random numbers and their correct sum.
- **Screenshots:**
 - Each screenshot should only show one run of the program (input and output).
 - Text in screenshots must be at least 14 points in font size for easy readability without zooming in.
 - Only capture what is necessary to demonstrate your program's functionality.
- **Organization:**
 - The lab report should be well-organized and easy to follow.
 - Use proper formatting and clear labels for each run.
- **Professional appearance:**
 - The report should be neat, with good spacing and consistent font usage.
 - Use proper indentation and avoid clutter.

Programming Requirements (tutor.cpp):

- Your program should generate two **random numbers less than or equal to 1000**.
 - Display the two numbers vertically formatted for addition, e.g.:
 - 247
 - + 129
 - -----
 - The program should **pause** after displaying the problem, waiting for the student to press a key to reveal the answer.
 - When the student presses a key, display the correct sum in the same format, e.g.:
 - 247
 - + 129
 - -----
 - 376
 - You are encouraged to use functions or any programming structures you have learned so far to make your code clean and modular.
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Additional Notes:

- Make sure your code is properly **indented and commented** where necessary.
 - Ensure your program handles multiple runs with different random numbers correctly.
 - Follow good coding practices to make your program easy to read and maintain.
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Summary:

File Name	Content
lab03.pdf	Lab report with screenshots of 5+ runs proving program correctness
tutor.cpp	Source code of your math tutor program

If you follow these guidelines, your submission will be complete, clear, and professional. Good luck!

Any question? Just email me.
LDV.